# Embed the widget



Learn how to add the Turnstile widget to your webpage using implicit or explicit rendering methods.

Turnstile offers two ways to add widgets to your page. **Implicit rendering** automatically scans your HTML for widget containers when the page loads. **Explicit rendering** gives you programmatic control to create widgets at any time using JavaScript. Use implicit rendering for static pages where forms exist at page load. Use explicit rendering for dynamic content and single-page applications (SPAs) where forms are created after the initial page load.

| Feature | Implicit rendering | Explicit rendering |
| --- | --- | --- |
| **Ease of setup** | Simple, minimal code | Requires additional JavaScript |
| **Control over timing** | Renders automatically on page load | Full control over rendering timing |
| **Use cases** | Static content | Dynamic or interactive content |
| **Customization** | Limited to HTML attributes | Extensive via JavaScript API |

## Prerequisites

Before you begin, you must have:

- A Cloudflare account
- A Turnstile widget with a sitekey
- Access to edit your website's HTML
- Basic knowledge of HTML and JavaScript

## Process

1. Page load: The Turnstile script loads and scans for elements or waits for programmatic calls.
2. Widget rendering: Widgets are created and begin running challenges.
3. Token generation: When a challenge is completed, a token is generated.
4. Form integration: The token is made available via callbacks or hidden form fields.
5. Server validation: Your server receives the token and validates it using the Siteverify API.

# Implicit rendering

Implicit rendering automatically scans your HTML for elements with the `cf-turnstile` class and renders widgets without additional JavaScript code. This set up is ideal for static pages where you want the widget to load immediately when the page loads.

## Use cases

Cloudflare recommends using implicit rendering on the following scenarios:

- You have simple implementations and want a quick integration.
- You have static websites with straightforward forms.
- You want widgets to appear immediately on pageload.
- You do not need programmatic control of the widget.

## Implementation

### 1. Add the Turnstile script

**Include the Turnstile Script**: Add the Turnstile JavaScript API to your HTML file within the `<head>` section or just before the closing `</body>` tag.

```
<script
  src="https://challenges.cloudflare.com/turnstile/v0/api.js"
  async
  defer
></script>
```

> **Warning**
> The `api.js` file must be fetched from the exact URL shown above. Proxying or caching this file will cause Turnstile to fail when future updates are released.

### 2. (Optional) Optimize performance with resource hints

Add resource hints to improve loading performance by establishing early connections to Cloudflare servers. Place this `<link>` tag in your HTML `<head>` section before the Turnstile script.

```
<link rel="preconnect" href="https://challenges.cloudflare.com" />
```

### 3. Add widget elements

Add widget containers where you want the challenges to appear on your website.

```
<div class="cf-turnstile" data-sitekey="<YOUR-SITE-KEY>"></div>
```

### 4. Configure with data attributes

Customize your widgets using data attributes. Insert a `div` element where you want the widget to appear.

```
<div
  class="cf-turnstile"
  data-sitekey="<YOUR-SITE-KEY>"
  data-theme="light"
  data-size="normal"
  data-callback="onSuccess"
></div>
```

Once a challenge has been solved, a token is passed to the success callback. This token must be validated against our Siteverify endpoint.

## Complete implicit rendering examples by use case

> **Basic login form**

> **Advanced form with callbacks**

> **Multiple widgets with different configurations**

> **Automatic form integration**

# Explicit rendering

Explicit rendering gives you programmatic control over when and where the widget appears and how the widgets are created using JavaScript functions. This method is suitable for dynamic content, single-page applications (SPAs), or conditional rendering based on user interactions.

## Use cases

Cloudflare recommends using explicit rendering on the following scenarios:

- You have dynamic websites and single-page applications (SPAs).
- You need to control the timing of widget creation.
- You want to conditionally render the widget based on visitor interactions.
- You want multiple widgets with different configurations.
- You have complex applications requiring widget lifecycle management.

## Implementation

### 1. Add the script to your website with explicit rendering

```
<script
  src="https://challenges.cloudflare.com/turnstile/v0/api.js?render=explicit"
  defer
></script>
```

### 2. Create container elements

Create containers without the `cf-turnstile` class.

```
<div id="turnstile-container"></div>
```

### 3. Render the widgets programmatically

Call `turnstile.render()` when you are ready to create the widget.

JavaScript

```javascript
const widgetId = turnstile.render("#turnstile-container", {
  sitekey: "<YOUR-SITE-KEY>",
  callback: function (token) {
    console.log("Success:", token);
  },
});
```

## Optional calls

After rendering the Turnstile widget explicitly, you may need to interact with it based on your application's requirements. Refer to the sections below to manage the widget's state.

### Reset a widget

To reset the widget if the given widget timed out or expired, you can use the function:

JavaScript

```javascript
turnstile.reset(widgetId);
```

### Get the response token

Retrieve the current response token at any time:

JavaScript

```javascript
const responseToken = turnstile.getResponse(widgetId);
```

### Remove a widget

When a widget is no longer needed, it can be removed from the page using:

JavaScript

```javascript
turnstile.remove(widgetId);
```

This will not call any callback and will remove all related DOM elements.

## Complete explicit rendering examples by use case

> Basic explicit implementation

> Using onload callback

> Advanced SPA implementation

## Widget lifecycle management

Explicit rendering provides full control over the widget lifecycle.

JavaScript

```javascript
// Render a widget
const widgetId = turnstile.render("#container", {
  sitekey: "<YOUR-SITE-KEY>",
  callback: handleSuccess,
});

// Get the current token
const token = turnstile.getResponse(widgetId);

// Check if widget is expired
const isExpired = turnstile.isExpired(widgetId);

// Reset the widget (clears current state)
turnstile.reset(widgetId);

// Remove the widget completely
turnstile.remove(widgetId);
```

## Execution mode

Control when challenges run with execution modes.

JavaScript

```javascript
// Render widget but don't run challenge yet
const widgetId = turnstile.render("#container", {
  sitekey: "<YOUR-SITE-KEY>",
  execution: "execute", // Don't auto-execute
});

// Later, run the challenge when needed
turnstile.execute("#container");
```

## Performance and user experience optimization

Cloudflare recommends that you execute the Turnstile script as early upon the visitor's page entry as possible, so that the verification is complete and the interaction is available once the visitor attempts an action on the page.

## Configuration options

Both implicit and explicit rendering methods support the same configuration options. Refer to the table below for the most commonly used configurations.

| Option | Description | Values |
| --- | --- | --- |
| `sitekey` | Your widget's sitekey | Required string |
| `theme` | Visual theme | `auto`, `light`, `dark` |
| `size` | Widget size | `normal`, `flexible`, `compact` |
| `callback` | Success callback | Function |
| `error-callback` | Error callback | Function |
| `execution` | When to run the challenge | `render`, `execute` |
| `appearance` | When the widget is visible | `always`, `execute`, `interaction-only` |

For a complete list of configuration options, refer to Widget configurations.

# Testing

You can test your Turnstile widget on your webpage without triggering an actual Cloudflare Challenge by using a testing sitekey.

Refer to Testing for more information.

# Limitations

Turnstile is designed to function only on pages using `http://` or `https://` URI schemes. Other protocols, such as `file://`, are not supported for embedding the widget.

# Security requirements

- Server-side validation is mandatory. It is critical to enforce Turnstile tokens with the Siteverify API. The Turnstile token could be invalid, expired, or already redeemed. Not verifying the token will leave major vulnerabilities in your implementation. You must call Siteverify to complete your Turnstile configuration. Otherwise, it is incomplete and will result in zeroes for token validation when viewing your metrics in Turnstile Analytics.
- Tokens expire after 300 seconds (5 minutes). Each token can only be validated once. Expired or used tokens must be replaced with fresh challenges.

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